



PaN-Finder Deliverable 3.c:

Revised System Benchmarks

Introduction

The PaN-Finder project has developed a benchmark system tailored to test and measure the performance and accuracy of the system and its design and components.

Every time a change is made to the system, a new run of the benchmark system is performed. A benchmark run consists of a number of settings (called hyper parameters) applied to the combined system that comprises the application and the benchmark itself, all the tests performed, the responses of the application and a set of metrics that are computed from the results and additional statistics collected.

In short, a run can be indicated as the following function:

$$Rn_z = R(B_i, A_j, H_v) = (Rs_z, M_z)$$

where

- Rn_z is benchmark run z
- B_i is benchmark version i
- A_j is application version j
- H_v is hyper parameters version v
- Rs_z is all the results set returned by the application
- M_z is all the metrics values collected and computed from the results sets R_v

The hyper parameters is a tuple of sets defined as follow:

$$H = (K, N, T, M)$$

where:

- H is the hyper parameters selected
- K are the K values used for the ranking and scoring
- N is the number of items returned in the results set at each call.
- T is the set with the test pairs selected
- M is the set of selected metrics. Some of the metrics are applied to each individual test pair, some of them are computed at the run level



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Each version of the application A_j will be tested with multiple benchmark runs with a selected combination of the hyper parameters values. The hyper parameters values will be explained in the following pages.

Each test pair is a three elements tuple as follows:

$$TP = (P_u, I_T, r_e)$$

where:

- P_u is the user prompt
- I_T is the target item or items, aka the item or items that we are testing against. The target items are indicated by their DOI.
- r_e is the expected ranking of the item. This is the position of the item in the list of results set returned. Initially it is always set to 1 (top position), as we do not have any prior knowledge regarding the topic. We are hopeful to develop such knowledge as we collect more benchmarks runs

The results of each benchmark run together with the definition of each test pairs are contained in a dedicated MongoDB database.

Each test pair is saved in the benchmark database under the collection *test_pairs* with the following schema:

- tpId UUID PRIMARY KEY,
- userPrompt TEXT,
- targetDoi TEXT,
- expectedRank INTEGER,
- promptID UUID,
- expertName TEXT,
- groupId TEXT,
- type test_pair_type,
- created DATETIME DEFAULT CURRENT_TIMESTAMP,
- deprecated DATETIME,

When a run is started, the benchmark code will load the selected test pairs indicated in the hyper parameters, submit each one of them through the application, collect the results sets, together with execution times, and compute the individual test pairs metrics. Once all the test pairs have been submitted, the run metrics are computed. All this information is be saved in the database, in order to be able to compare different hyper parameters set and different version of the application.

The benchmark run will be saved in the table *benchmarks_run* with the following schema:

- runId UUID PRIMARY KEY,
- startedAt DATETIME DEFAULT CURRENT_TIMESTAMP,
- endedAt DATETIME DEFAULT CURRENT_TIMESTAMP,



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- hyperparameters: JSONB

The test of the individual test pair is saved in the table named *benchmarks_run_test* with schema:

- testId UUID PRIMARY,
- runId UUID REFERENCES benchmarks_run(runId),
- tpId UUID REFERENCES test_pairs(tpId),
- intention TEXT,
- keywords TEXT[],
- filters JSONB,
- rank INTEGER,
- runTime decimal,
- extractionTime decimal,
- buildingTime decimal,
- queryTime decimal,
- sqlQuery TEXT,
- resultSet JSONB

The metrics calculated for each test pair are stored in table *test_pair_metric_values* with schema:

- id SERIAL PRIMARY KEY,
- runId UUID REFERENCES benchmarks_run(runId),
- testId UUID REFERENCES benchmarks_run(testId),
- metric TEXT,
- value DECIMAL

While the metrics calculated at the run level are stored in *run_metrics_values* with schema:

- id SERIAL PRIMARY KEY,
- runId UUID REFERENCES benchmarks_run(runId),
- metric TEXT,
- value DECIMAL

Test Pairs

Originally, we planned to include in the PaN-Finder benchmark only *Single Positive* test pairs, which are defined as:

- Given a single user query, we expect that the results set contains the specified item.

Based on our initial interaction with the system, we assessed that the benchmark should be expanded to include also the following test pairs:

- *Single Negative*:
Given a single user query, we expect that the results set does not contains the specified a item.



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- *Multiple Positive:*

Given a single user query, we expect that the results set contains the specified items, which can be two or more.

With further knowledge of the system behavior and additional considerations, we decided to stop the implementation of Single Negative and Multiple Positive tests pair, at least at this point in time. We opted to introduce new test pair types which requires a dedicated matching metric. The new test pair types are:

- Extended
- Comparative

We will described these test pair types in the related extended metrics section below.

Metrics

The metrics can be grouped in two overall categories: test pair metrics and run metrics. The test pair metrics are computed on the individual test pair, while the run metrics are computed on the whole run computing the average of the matching test pair metrics across all test pairs used in the run.

The test pair metrics, which are computed on the individual test pair, are depended on the type of test pair.

Before diving in the definition of the PaN-Finder metrics, we first introduce the general metrics as they are found in the current literature. We will proceed to describe the custom version of the different metrics that have been adapted to our use case.

General Metrics

This section introduces different metrics that are applicable to LLM and information retrieval systems. Some of the metrics listed below are commonly used in the industry and well defined the current available literature. Some of them have been defined within the scope of this project, in order to better benchmark the project and its accuracy, and also uncover all the the subtle changes to all the tweaks applicable to the system.

- Precision @ K (Pc@K)

Measures how many items with the top K positions are relevant. This metrics does not have rank awareness.

$$Pc @ K = \frac{N_{r@K}}{K}$$

where

- $N_{r@K}$ = Number of relevant items in the first K items of the results set
- K = number of best ranked items considered in the results set



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- Context Precision @ K (CP@K)

Context Precision is a metric that evaluates whether all of the relevant items present in the results are ranked higher or not. Ideally all the relevant items must appear at the top ranks. This metrics has rank awareness, so the closer are the relevant items to the top, the higher is the score.

It's definition is:

$$CP@K = \frac{1}{N_{r@K}} \sum_{k=1}^K PC@k * v_k$$

where

1. $PC@k$ = Precision @ k
2. v_k = relevance indicator at rank k. It is 1 if item at rank k is relevant, 0 otherwise.
3. $N_{r@K}$ = Number of relevant items in the top K items of the result set
4. K = number of best ranked items considered in the results set,

- Reciprocal rank at K, RR@K

The reciprocal rank of a query response is the multiplicative inverse of the rank of the first correct answer.

$$RR@K = \frac{1}{r_1}$$

where

- r_1 = rank of the first relevant item.

- Presence@K, Ps@K

This metric has been defined within the scope of the PaN-Finder project. The advantages of this metric are the fact that it is easy to compute and intuitive to understand.

This metric indicate if in the top K items there is at least one relevant item. It can be calculated using Precision @ K. This metric does not have rank awareness. Its definition is:

$$Ps@K = \begin{cases} 1 & \text{if } PC@K > 0 \\ 0 & \text{otherwise} \end{cases}$$

- Linear Zero Position @ K (lzp@K)

This metrics average the weight across the number of relevant items in the top K items of the results set. For each relevant item the metric assign a weight from 1 to 0 that linearly decrease with the rank. It zeros out at K+1. Relevant items that are at the end of the list are assigned a



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lower weight. This metric has rank awareness.

$$lzp @ K = \frac{1}{N_{r@K}} \sum_{k=1}^K \left(1 - \frac{k-1}{K}\right) * v_k$$

where

5. $N_{r@K}$ = number of relevant items in the results set
6. K = selected number of top items
7. V_k = relevance indicator at rank k . It is 1 if item at rank k is relevant, 0 otherwise.

- Exponential Zero Position @ K ($ezp@K$)

This metrics average the weight across the number of relevant items in the top K items of the results set. For each relevant item the metric assign a weight from 1 to 0 that exponentially decrease with the rank. It zeros out at $K+1$. This metric has rank awareness.

$$ezp @ N = \frac{1}{N_{r@K}} \sum_{k=1}^K \left(\frac{e^{\frac{1}{S}(K-k)} - 1}{e^{\frac{K}{S}} - 1} \right) * v_k$$

where

8. $N_{r@K}$ = number of relevant items in the results set
9. K = number of selected top items in the results set
10. S = Curvature Scaling Constant with $S \in \mathbb{Z}_+^*$. Suggested value 1/3.
11. V_k = relevance indicator at rank k . It is 1 if item at rank k is relevant, 0 otherwise.

Metrics with 1 relevant item

The majority of the current test pairs contain only one relevant item, associated with each test query. Based of this condition, we can rewrite the metrics presented in the previous section and simplify their definitions as follow.

- Precision @ 1K ($Pc@1K$)

Precision at 1K, is derived from $Pc@K$ substituting $N_{r@K}$ with 1 if the target item is present in the top K results, 0 otherwise.

The definition for Precision at 1K is:



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$$Pc@1K = \begin{cases} \frac{1}{K} & \text{if } 0 < r \leq K \leq N_{rs} \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the relevant item in the results set
- N_{rs} = number of elements in the results set
- K = number of top results considered in the results set

With the current definition, this metric is not relevant for our use case. It is equivalent to and can be replaced by Presence@1K, which is easier to compute.

- Context Precision @ 1K (CP@1K)

Substituting the relevant information in the CP@K, we can define the CP@1K as:

$$CP@1K = \begin{cases} \frac{1}{r} & \text{if } 0 < r \leq K \leq N_{rs} \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the relevant item in the results set
- N_{rs} = Number of items in the results set
- K = number of top items considered in the results set,

- Discounted Cumulative Gain

- Presence @ 1K (Ps@1K)

Given that we have only one relevant result, the Ps@1K definition becomes:

$$Ps@1K = \begin{cases} 1 & \text{if } 0 < r \leq K \leq N_{rs} \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the relevant item in the results set
- N_{rs} = number of items in the results set
- K = number of top items considered in the results set,

- Linear Zero Position @ 1K, lzp@1K

This metrics assign a value from 1 to 0 that linearly decrease with the rank. It zeros out at K+1. This metric has rank awareness similar to CP@1K. Reworking the definition of lzp@K with the



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knowledge of the 1 relevant item, its definition is:

$$lzp@1K = \begin{cases} 1 - \frac{r-1}{K} & \text{if } 0 < r \leq K \leq N_{rs} \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the relevant item in the results set
- N_{rs} = number of items in the results set
- K = selected number of top items

8. Exponential Zero Position @ 1K, [ezp@1K](#)

This metrics assign a value from 1 to 0 that exponentially decrease with the rank. It zeros out at $K+1$. This metric has rank awareness similar to [CP@1K](#) and its definition is:

$$ezp@1N = \begin{cases} \frac{e^{\frac{1}{S}(K-r)} - 1}{e^{\frac{K}{S}} - 1} & \text{if } 0 < r \leq K \leq N_{rs} \\ 0 & \text{otherwise} \end{cases}$$

where

1. r = rank of the desired document in the results set
2. N_{rs} = number of items in the results set
3. K = number of selected top items in the results set
4. S = Curvature Scaling Constant with $S \in \mathbb{Z}_+^*$. Suggested value 1/3.

Single Positive Test pair metrics

One of the test pair types used in the PaN-Finder benchmarks are the single positive test pairs, which contains one answer, or relevant item, that we expect in the results set, matched with a specific user prompt. This additional condition leads us to rewrite the metrics for a single relevant item, or @1K where we can substitute K with the number of items in the results set N_{rs} , or simply N .

The metrics computed on each single positive test pairs are the following:

- Context Precision @ 1N (CP@1N)

This is context precision at the arity of the results set with at most 1 relevant item.

This metrics is also equivalent to the Reciprocal Rank @ N , $RR@N$, with 1 relevant item

$$CP@1N = \begin{cases} \frac{1}{r} & \text{if } 0 < r \leq N \\ 0 & \text{otherwise} \end{cases}$$



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where

- r = rank of the relevant item in the results set
- N = Number of items in the results set

- Presence @ 1N ($Ps@1N$)

This metric indicate if the relevant item is present in the results set.

$$Ps@1N = \begin{cases} 1 & \text{if } 0 < r \leq N \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the relevant item in the results set
- N = Number of items in the results set

- Linear Zero Position @ 1N ($lzp@1N$)

$$lzp@1N = \begin{cases} 1 - \frac{r-1}{N} & \text{if } 0 < r \leq N \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the relevant item in the results set
- N = number of items in the results set

- Exponential Zero Position @ 1N ($ezp@1N$)

$$ezp@1N = \begin{cases} \frac{e^{\frac{1}{S}(N-r)} - 1}{e^{\frac{N}{S}} - 1} & \text{if } 0 < r \leq N \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the desired item in the results set
- N = number of items in the results set
- S = Curvature Scaling Constant with $S \in \mathbb{Z}_+^*$. Suggested value 1/3.

- Linear Expected Position @ 1N ($lep@1N$)

This metrics assigns a higher linear value between 0 and 1 if the relevant item is at closer to the expected position in the list.



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At the moment, this metric is not relevant in the benchmarks as we do not have a definitive knowledge of which is the expected position of the relevant item in the results set. We are hoping to be able to develop such knowledge, increase the meaningfulness of this metric and include it in our evaluation of the system.

The formal definition of the metric is:

$$lep@1N = \begin{cases} \frac{1}{N} |r_e - r| & \text{if } 0 < (r_e, r) \leq N \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the desired item in the results set
- N = number of items returned in the results set
- r_e = expected rank provided by domain knowledge experts or previous benchmarks

- Exponential Expected Position @ 1N (eep@1N)

This metrics assigns a higher exponential value between 0 and 1 if the relevant item is closer to the expected position in the list.

The same consideration highlighted for lep@1N are applicable for this metrics.

$$eep@1N = \begin{cases} \frac{e^{\frac{1}{S}|r_e - r|} - 1}{e^{\frac{N}{S}} - 1} & \text{if } 0 < r \leq N \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the desired item in the results set
- N = number of items returned in the results set
- r_e = expected rank provided by domain knowledge experts or previous benchmarks
- S = Curvature Scaling Constant with $S \in \mathbb{Z}_+^*$. Suggested value 1/3.

- zero normalized Discounted Cumulative Gain @ 1N (zNDCG@1N)

This is a ranking quality metric. It compares rankings to an ideal order where all relevant items are at the top of the list.

$$znDCG@1N = \begin{cases} \frac{1}{\log_2(r+1)} & \text{if } 0 < r \leq N \\ 0 & \text{otherwise} \end{cases}$$

where



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- r = rank of the desired item in the results set
- N = number of items returned in the results set

1. expected normalized Discounted Cumulative Gain @ 1N ($eNDCG@1N$)

$$eNDCG@1N = \begin{cases} \frac{\log_2(r_e+1)}{\log_2(r+1)} & \text{if } 0 < r \leq N \\ 0 & \text{otherwise} \end{cases}$$

where

- r = rank of the desired item in the results set
- N = number of items returned in the results set
- r_e = expected rank provided by domain knowledge experts or previous benchmarks

Single Negative Test Pairs Metrics

In the PaN-Finder benchmarks, we are planning to include single negative test pairs, which are tuple with a query and the reference to an item that should not be in the results set.

The only metric that makes sense in this case is the inverse of the Presence @ K, which we call Absence @ K:

1. Absence @ 1N ($Ps@1N$)

This metric indicate if the relevant item is not present in the results set.

$$Ps@1N = \begin{cases} 0 & \text{if } 0 < r \leq N \\ 1 & \text{otherwise} \end{cases}$$

where

- r = rank of the relevant item in the results set
- N = Number of items in the results set

Extended Test Pairs Metrics, round 1

After the soft lunch of the PaN-Finder and the introduction of the knee-point algorithm to differentiate between relevant and suggested results, we reviewed the user queries submitted and further interacted with the tool. This lead to a brainstorm session and reached the conclusion that basic metrics and basic test pairs were still useful but not enough to cover some use cases.

We decided to introduce additional extended metrics which takes into account if, given a specific query, the document that we are looking for should be a relevant, suggested or should not be present at all (negative test). Further more, we have discovered use cases when the user provides information about a



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query and two or more documents that should be presented in a specific order relative to each other. Each new metrics can be computed if the extended matching test pair is provided. The new test pair will be indicated together with the metric definition.

1. **Extended Presence @ 1N** (EP@1N)

This metric indicate if the relevant item is in the correct group indicated in the test pair. The metric provides a score based on the weight specified in the definition.

The formal metric is defined as:

$$EP @ 1 N = w_{Ga, Gb}$$

where:

- $S_{Ga, Gb}$ is the defined scored for the G_a, G_b pair
- G_a is the actual group
- G_e is the expected group

This metric requires the definition of the score metrics which assign the score for each case, such the score if the item was supposed to be in the HR group but was listed in the LR one or was no present at all.

One example of EP score matrix can be the following:

Expected	Actual			
		HR	LR	NP
	HR	1	0.25	-1
	LR	0.25	1	-1
	NP	-1	-1	1

The score matrix is part of the parameter space defined in the benchmark hyperparameter files.

The test pair associated with this metric is defined as following:

(t_i, q_i, doi_i, g_i)

Where:

- t_i is the type of the test pair, in this case EP
- q_i is the query submitted to the system



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- doi_i is the document that we are targeting
- g_i is the group of this document should belong to which can have one of the following values:
 1. HR = High Relevance. In the FE, it is shown as a relevant document
 2. LR = Low Relevance. In the FE, it is shown as a suggested document
 3. NP = Not Present. The document should not in the result set

2. **Comparative Presence @ N (CP@N)**

This metric provide a score which indicate if the listed items are in the correct relative position compared to each other.

The metric is defined as:

$$CP@N = \frac{1}{N} \sum_{i=1}^{N-1} \begin{cases} 1 & \text{if } r_{di} > r_{d(i+1)} \\ 0 & \text{otherwise} \end{cases}$$

where:

1. R_{di} is the rank of document i
2. R_{d+1} is the rank of document $i+1$
3. N is the number of document listed

The test pair associated with this metric is defined as following:

$(t_i, q_i, doi_{i1}, doi_{j2}, \dots, doi_{jN})$

Where:

- i is the i test pair
- t_i is the type of the test pair, in this case CP
- q_i is the query submitted to the system
- doi_{ij} is the document that should come at the j position in comparison to the other documents
- N is the number of the targeted documents

Time metrics

Additional test pairs metrics are the time metrics, which are saved directly in the test pair test record:



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1. Total Execution Time, TET
2. Prompt Extraction Time, PET
3. Query Building Time, QBT
4. Sql Query Time, SQT

Run metrics

A run consist of the tuple including K values of the score computation, the number of items returned (used as the @K in the metrics) and the set of test pairs, which can be all them (named comprehensive), only synthetic (AI generated) or expert (provided by a domain expert or an informed user). All this values are considered and named hyper-parameters of the benchmark.

For each run, we compute the metrics specified in the test pair section and, with them, we compute the mean of the values computed with the run.

The metrics for the run are defined as follow:

$$\overline{M}(H) = \overline{M}(T, N) = \frac{1}{\text{card}(T)} \sum_{t \in T} M @ N(t)$$

where:

1. M is the metric considered
2. H are the hyper-parameters of the run
3. T is the set of test pairs considered
4. N is the number of items returned in the results set

The following table lists all the metrics that should be computed for each run, after the individual test pair metrics are computed. Some hyper parameters, including N, are not set yet as their best value is still under investigation.

Given the hyper parameters, the run metric will be named according the following naming convention:

tMmm

where

1. *t* is the test pair sets used for the run metric. At the time of this writing, the possible options are: c for comprehensive, which includes all the single positive test pairs, s for synthetic, which includes only the synthetic pairs AI generated, and e which contains all the test pairs provided by real users and expert domains.
2. *mm* is the selected test pair metric, which can be any of the following: CP, Ps, lzp, ezp, lep, eep, zNDCG, or eNDCG.

The extended name of the run metric should actually be:

tMmm@n(h)



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where:

3. n is the number of items returned (or arity) of the results set
4. h is the rest of the hyper parameters that are not used directly in the calculation of the metric but are important in the computation of the score. In the database, this portion is omitted as the record is already connected to run record which contains the information

but n and h values are implicit with the specific run, so we simplify the naming.

Table of all the run metrics provided

Test Pairs Set			Name	Comprehensive	Synthetic	Expert
Test Pairs Metric name			Symbol	T_C	T_S	T_E
Full	Short	Selected				
Context Precision	CP	y	MCP	cMCP	sMCP	eMCP
Presence	Ps	y	MPs	cMPs	sMPs	eMPs
Linear zero position	lzp	n	MLzp	cMLzp	sMLzp	eMLzp
exponential zero position	ezp	n	Mezp	cMezp	sMezp	eMezp
Linear expected position	lep	n	Mlep	cMlep	sMlep	eMlep
Exponential expected position	eep	n	Meep	cMeep	sMeep	eMeep
zero normalized Discounted Cumulative Gain	znDCG	n	MznDCG	cMznDCG	sMznDCG	eMznDCG
expected normalized Discounted Cumulative Gain	enDCG	n	MenDCG	cMenDCG	sMenDCG	eMenDCG
Extended Presence	EP	y^*	MEP	cMEP	sMEP	eMEP
Comparative Presence	CP	y^*	MCP	cMCP	sMCP	eMCP

At the moment, we are not able to prove if all of the above-mentioned metrics are valuable and informative for our use case. Our approach is to initially compute all of them until we can make an informed decision, based on the results, about which one to keep and to drop.



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In addition to the run metrics mentioned in the table above, we will also compute the mean execution time for each run, which is dependent from the hyper parameters set for the run itself. Here is the list of run time metrics:

1. Mean Total Execution Time, MTET@K
2. Mean Prompt Execution Time, MPET@K
3. Mean Sql Execution Time, MSET@K

Resources

1. <https://medium.com/@med.el.harchaoui/rag-evaluation-metrics-explained-a-complete-guide-dbd7a3b571a8>
2. <https://medium.com/@plthiyagu/comprehensive-evaluation-metrics-for-retrieval-augmented-generation-rag-a846ec355c86>
3. <https://www.confident-ai.com/blog/rag-evaluation-metrics-answer-relevancy-faithfulness-and-more>
4. <https://www.confident-ai.com/blog/the-ultimate-llm-evaluation-playbook>
5. <https://www.confident-ai.com/blog/llm-evaluation-metrics-everything-you-need-for-llm-evaluation>
6. <https://www.evidentlyai.com/ranking-metrics/precision-recall-at-k>
7. https://en.wikipedia.org/wiki/Discounted_cumulative_gain
8. <https://www.evidentlyai.com/ranking-metrics/ndcg-metric>
9. <https://www.evidentlyai.com/ranking-metrics/precision-recall-at-k>
10. https://docs.ragas.io/en/stable/concepts/metrics/available_metrics/context_precision/
11. <https://oecd.ai/en/catalogue/metrics/context-precision-1>
12. https://en.wikipedia.org/wiki/Mean_reciprocal_rank



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